

DentAR: Innovating Dental Visits with Sensory Experiences in AR for People with Autism Spectrum Disorder

Ge (Serena) Guo
Information Science,
Cornell University
Ithaca, New York, USA
gg372@cornell.edu

Zhiwen Qiu
Information Science,
Cornell University
Ithaca, New York, USA
zq76@cornell.edu

ABSTRACT

In dental clinics, individuals with Autism Spectrum Disorder (ASD) often face heightened anxiety and discomfort due to Sensory Processing Disorders (SPD), exacerbated by the clinical environment. This study introduces *DentAR*, an Augmented Reality (AR) environment designed to mitigate SPD-related anxiety in dental settings for individuals with ASD. Developed via a participatory design approach with occupational therapists, dentists, and caregivers, *DentAR* utilized the sensory integration strategy to modify the sensory characteristics of the dental clinic, by employing visual and auditory stimuli to enhance the dental experience for those with SPD. Preliminary expert evaluations suggest potential benefits in using AR for sensory integration, emphasizing the importance of personalization and user empowerment in the design of therapeutic interventions. This research demonstrates AR's utility in addressing sensory processing challenges in clinical environments, to facilitate accessible therapeutic interventions for diverse sensory needs.

CCS CONCEPTS

• **Human-centered computing** → **Field studies; Ubiquitous and mobile computing design and evaluation methods.**

KEYWORDS

ASD, SPD, Augmented Reality (AR), behavior intervention, context, dental clinic

ACM Reference Format:

Ge (Serena) Guo and Zhiwen Qiu. 2024. DentAR: Innovating Dental Visits with Sensory Experiences in AR for People with Autism Spectrum Disorder. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3650926>

1 INTRODUCTION

Sensory abnormalities are reported to be central to the autistic experience. Studies have found that over 90% of adults with Autism Spectrum Disorder (ASD) had sensory abnormalities [9, 16] within

different sensory processing domains, particularly to tactile, auditory, and visual stimuli. The symptoms may vary from hyposensitivity, hypersensitivity, or difficulty with multi-sensory integration [12].

Sensory Processing Disorder (SPD) is particularly critical in clinical settings, especially in dental clinics [28]. Individuals with SPD may have an exaggerated response to sensory stimuli, such as the sound of a dental drill, the bright lights in the dental office, or the sensation of dental tools in their mouth. This heightened sensitivity can lead to anxiety, fear, and discomfort, making routine dental procedures more challenging [7]. It not only exacerbates the stress associated with dental visits but also poses a barrier to essential oral healthcare. Research has shown that people with SPD are particularly vulnerable to oral health disparities, significantly impacting their overall well-being [11].

Many therapies have been developed to address the sensory abnormality of SPD, such as sensory-integration training [26]. However, due to the limited number and availability of experienced therapists, providing consistent interventions poses significant challenges. Consequently, there's a growing need for more effective, context-based therapeutic approaches. Technological interventions have emerged as a possible solution, offering low-cost and widely accessible therapeutic options that are not constrained by the availability of professional therapists. For example, *MEDIATE* [29] is a multi-sensory, adaptive environment that utilizes visual, auditory, and tactile stimuli to enhance sensory processing and interaction in people with SPD.

Another emerging technology is Augmented Reality (AR), which blends computer-generated elements into the real world, allowing people to experience the physical environment overlaid with computer graphics [18]. AR as a spatial technology has the potential to augment our surrounding environment by integrating sensory experiences, especially in the visual and auditory domains. This is especially beneficial for individuals with sensory challenges in the physical world. Recent approaches in AR for autism have explored different aspects such as social skill training [8], language training [23], emotion recognition [15], etc. However, there are a limited number of studies that consider using AR specifically to target the sensory integration approach in our physical environment, especially for people with SPD.

Our study investigated *DentAR*, an AR-based environment designed to alleviate anxiety for SPD sufferers in clinical settings. Developed in collaboration with a dental clinic and a non-profit organization, our participatory design approach [27] included occupational therapists (OTs), dentists, behavioral analysts, caregivers, and patients, from the beginning of the project. *DentAR* integrated

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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650926>

visual and auditory stimuli to transform tactile interactions during dental visits, aiming to reduce SPD-related anxiety. An initial evaluation involving four OTs, two dentists, and a behavior analyst, has demonstrated the interface's potential to enhance patients' comfort.

2 RELATED WORK

2.1 Enhancing Sensory Processing Through Sensory Integration

In psychology, OTs have utilized sensory-oriented interventions, such as music therapy, weighted blankets, swings, brushing, or massage, to improve sensory comfort in patients and as an effective training approach [17]. However, these treatments typically focus on one sensory modality.

Sensory integration treatment, with a focus on multiple senses as an integrated experience, is a highly unitized intervention in autism with SPD [26]. This therapeutic approach seeks to enhance those with SPD to effectively process and react to sensory inputs from their surroundings. It includes engaging in activities specifically designed to stimulate or distract the individual's sensory processing [14]. These activities are often playful in nature and are geared towards fostering their motor skills, balance, and coordination, while also providing appropriate sensory stimuli to aid in better organization of their nervous systems [14, 25].

Drawing on these insights, recent research has highlighted the success of sensory-integrated environments in enhancing sensory processing. For instance, one study demonstrated the effectiveness and feasibility of a Sensory-Adapted Dental Environment (SADE), with modified visual, auditory, and tactile stimuli in dental clinics [7]. This innovation has led to reduced physiological stress, anxiety, behavioral distress, and sensory discomfort for children with ASD during dental cleanings [7]. Another research involving 162 autistic children exhibited significantly lower physiological stress in SADE compared to traditional environments [28].

In our work, we adopted the sensory-integration approach, utilizing AR to enrich the sensory experience in clinical contexts. This method transformed the physical environment to integrate visual and auditory stimuli, amplifying the therapeutic benefits for individuals with SPD.

2.2 Multi-sensory Environments in HCI

HCI research has explored the potential of multi-sensory environments in aiding individuals with sensory processing disorders, particularly among the autistic population. For example, *MEDIATE* [29] offered a multi-sensory, adaptive space aimed at fostering interactions and expression in individuals with ASD, particularly for those with limited verbal communication abilities. This environment incorporated visual, auditory, and tactile stimuli, adjusting dynamically in real-time to encourage engagement, exploration, and creativity in a safe and controlled setting. Building on this, Novakovic and colleagues introduced the *Snoezelen* environment [22], crafted to deliver sensory stimuli through lights, colors, sounds, and textures, to alleviate distress and enhance sensory processing and behavioral outcomes for those with SPD.

More recently, the field is progressing to leverage mixed reality technologies to address sensory processing challenges. Mixed reality environments offer unparalleled control over environmental

variables, enabling researchers and clinicians to tailor environments optimally for promoting learning and skill generalization. For instance, *Evenness* is a Virtual Reality (VR) environment designed to provide either calming or stimulating sensory input, thereby augmenting sensory processing, well-being, and adaptive behaviors for individuals with SPD [19]. Similarly, an AR project named *Starrypia* introduced a gamified multi-sensory intervention using music to keep children focused and calm [6]. These advancements underscored the potential of VR and AR technologies in devising novel interventions for individuals facing sensory processing challenges.

In our project, we have developed a contextualized AR experience that enables individuals with SPD to alleviate their anxiety in real-world contexts. To ensure a comprehensive understanding of their needs, we employed a participatory design process that involved various stakeholders including OTs, behavioral analysts, dentists, patients, and their caregivers in the design process.

3 DISCOVERY AND EXPLORATION PROCESS

This study, undertaken as part of a fellowship program, was conducted in partnership with a dental clinic serving the intellectual and developmental disabilities (I/DD) community. Our methodology was grounded in participatory design principles, involving collaboration with an interdisciplinary team of OTs, behavioral analysts, and dentists, all with significant experiences in assisting individuals with autism.

3.1 Stakeholder Interviews and Information Gathering

The initial phase of the research involved conducting focus group interviews with OTs and dentists to gain an in-depth understanding of their daily interactions with autistic patients, the challenges they encountered, and their strategies to facilitate dental visits for these individuals. These conversations helped us understand the sensory-oriented interventions currently implemented in clinics, elucidating a range of techniques aimed at reducing patients' fear and anxiety, while also delineating their limitations.

3.2 Observational Study and Patient Interaction

Another methodological approach in our research was the direct observation and active participation in dental appointments of patients with severe autism. Photos taken from one of these sessions, as shown in Figure 1, indicated that OTs were assisting a patient during their dental visit. These observation sessions were conducted after obtaining written consent from the patient's caregivers. We documented the dental visit procedures with video recordings, photos, and on-site notes. After each observational session, we conducted interviews with OTs to gain a deeper understanding of patient behaviors, and in collaboration with them, we have delineated the challenges encountered by patients throughout their dental appointments.

3.3 Identifying Three Challenges

Throughout this exploratory process, a primary challenge we identified was the patients' diverse sensory tolerances to environmental stimuli in the clinic. Specifically, there were three key challenges associated with tactile sensitivities during a dental visit: (A) The



Figure 1: Photos taken from one of the observational studies of the patient's dental visit sessions. In the photos, the Occupational Therapist (OT) was helping the patient going through different stages of the dental visit in the clinic. Consent forms for media release were acquired from the OT and the patient's caregiver.

experience of walking through the corridor, where the narrow and unfamiliar space could induce anxiety; (B) The approaching of the dental chair, which is fraught with the anticipation of tactile sensations from sitting down and the forthcoming unpleasant dental treatment procedures; (C) The interaction with dental tools, where the direct contact of these tools with their body could be particularly stressful.

3.4 Storyboarding and Low-Resolution Prototyping

After identifying the main challenges, we initiated the design process by creating storyboards for potential interaction scenarios to help us conceptualize ways to leverage various sensory inputs to divert patients' attention from their fears. Drawing inspirations from the *Snoezelen* design [4, 22], where rising water bubbles were used to induce calming sensations in individuals with sensory processing disorder, we incorporated bubble imagery in our AR system. To enhance the experience, we also added the sounds of rising bubbles in the water alongside the visuals. Figure 2 - a shows an excerpt from a storyboard depicting likely interactions in the dental room. Bubbles were activated in response to the patient's touch of the physical environment, such as when the patient stepped on the corridor floor, sat in the dental chair, or was contacted by dental tools. We employed techniques like hand sketching, Photoshop, and video-making to create and demonstrate the prototype's effect at a comparatively lower resolution. These methods facilitated rapid iterations of design solutions and ensured clear and on-time communication with various stakeholders. Figure 2 - b shows a preliminary iteration with Photoshop, by overlaying intended AR images on photos.

4 AR PROTOTYPE

4.1 Prototype Development

We developed the AR-assisted dental system using HoloKit AR Headset [3, 13], iPhone 14 Pro Max with iOS version 17.2.1 [2], and Unity3D graphics engine version 2021.3.33f1 [5]. The *DentAR* user interface was implemented with the support of the ARKit version 2.7.3 [1]. Designing the system requires three steps. First, we utilized ARKit's WorldMap function for spatial mapping, creating a detailed map of the clinical environment, including essential features such as surfaces, contours, and objects. The ARKit SDK was able to identify feature points in the environment to create reference points, known as "anchors", for accurate positioning of the overlaid virtual components. One of the key features of WorldMap was to persistently save and store these maps to enable a consistent AR experience in a real-world dental clinic setting across multiple sessions. These maps could be continuously updated in real-time to ensure that any changes in the environment were reflected in *DentAR* in a realistic and context-aware manner. Furthermore, the system incorporated occlusion dynamics, so that the virtual information was not presented when users' view was blocked by walls or body parts like hands and feet.

Second, we developed a multi-faceted interaction system to create an intuitive and engaging experience. This included interaction modalities to monitor users' positions, hand tracking, and real-time image detection for activation of virtual information with calming and relaxing sensory responses. Additionally, our system personalized the dental visit experience, catering specifically to the diverse and unique sensory needs of each individual, to ensure an accessible and inclusive dental environment. Finally, the *DentAR* system was deployed onto a smartphone and was fine-tuned for aligning AR content with the optics of the HoloKit head-mounted display. The incorporation of HMD emphasized a comfortable and hands-free

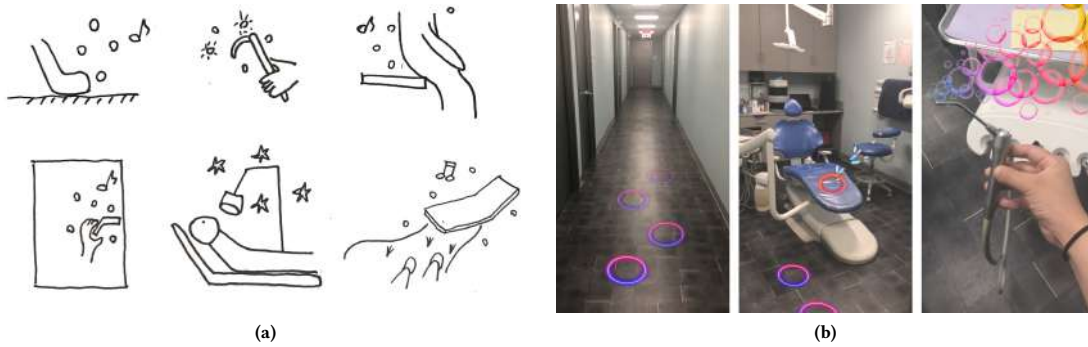


Figure 2: (a) An excerpt from Hand-sketch storyboards, and (b) Photoshop screenshots of the low-resolution prototype.

experience, enhancing the interactivity and control of the dental visit.

4.2 Design of Dental Experience

The overall goal of *DentAR* was to provide users with multisensory information to divert them from daunting aspects of dental visits while enhancing comfort and independence in clinical settings. We created three scenarios in the system: enhancing comfort in the clinic corridor, reducing anxiety with the dental chair, and alleviating the fear of dental tools. The system was designed to allow the patient's caregivers or OTs to fluidly determine the intensity of various sensory animations and audio to suit their sensory profiles.

4.2.1 Sensory Controls. In collaboration with the clinic, we discovered that each patient possesses a distinct sensory profile, established using a tool developed by Dunn [10], as depicted in Figure 3. This particular example, extracted from a case study of a patient with SPD [24], demonstrates the individual's exceptional sensitivity toward sounds. It further highlights the possible range of sensory sensitivities among patients, ranging from hypersensitivity to hyposensitivity towards external stimuli, such as light and sound. To accommodate these varied sensitivities, we designed Augmented Reality (AR) interfaces equipped with an adaptive sensory slider. This feature enabled caregivers and OTs to personalize the intensity of AR content, allowing adjustments in the size of virtual objects, speed of animations, and volume of accompanying audio. The system also integrated a voice command feature, enabling them to effortlessly fine-tune the intensity by commands such as "Lower" or "Higher". These features not only enhanced accessibility but also provided intuitive and seamless interactions.

4.2.2 Scenario 1 - Walking through the corridor. Users perceived corridor experience as excessively narrow and poorly illuminated, leading to heightened fear and a sense of diminished control of the environment. In response, we developed an ARKit-based feature that positioned multiple silver-colored rings on the ground by detecting horizontal planes of the corridor (Figure 4 - a). When individuals stepped onto one of these rings, interactive bubble animations with varying colors, sizes, and spatial sounds would be triggered, playing in loops and disappearing after a short duration. This was achieved by continuously tracking the user's position,

calculating their proximity to the rings, and ensuring that only one animation was displayed when they approached a ring to prevent overwhelmed feelings. These visual effects played a crucial role in managing and regulating emotional responses of individuals with SPDs. The simple and non-intrusive rings acted not only as visual markers to guide users in unfamiliar environments, but also offered predictable interactions to comfort those who found unknown destinations stressful. The view of bubbles rising to the ceiling also induced a soothing and enjoyable experience for anxiety reduction.

4.2.3 Scenario 2 - Approaching the dental chair. Dental chairs with hard and cold surfaces, situated in a room with unfamiliar equipment, can be particularly discomforting for those with heightened tactile sensitivities. Individuals with SPDs often encounter environmental stress when they feel confined and lose control, unable to ground their feet while lying in chairs. To mitigate these challenges, we have created an experience on seating areas, allowing them to interact with virtual rings positioned on the chair (Figure 4 - b). The hand-tracking feature activated tranquilizing animations accompanying with relaxing music by detecting the time when users' hands virtually touched the ring within the required interaction distance. The integration of natural elements in the animations, such as serene landscapes or flying butterflies, was designed to enhance comfort for patients. These animations were not static but dynamic; they would continuously evolve and persist as long as users remained seated. This ongoing visual stimulation helped to maintain a calm environment, effectively diverting their attention away from any potential anxiety-inducing aspects of the dental treatment. The overall goal of this experience was to engage users in comfort-giving environments, thereby positively reinforcing their willingness to participate in future dental visits.

4.2.4 Scenario 3 - Interaction with dental tools. Individuals with SPDs frequently experienced intimidation of sharp and metallic dental equipment due to the high-pitched noises from drills, and the tactile sensations when these tools contacted their teeth. Our application implemented an image detection system that recognized specific markers or images affixed to dental tools. As soon as users' view identified a marker, the system generated virtual bubble effects that seamlessly appeared to emanate from the tip of the dental tools in operation (Figure 4 - c). The real-time detection could further

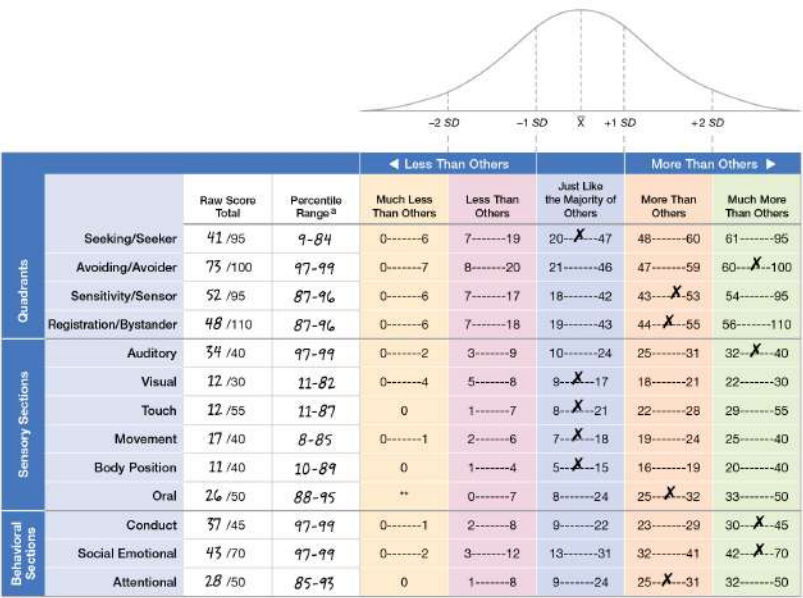


Figure 3: An illustrative example of a patient’s sensory profile.

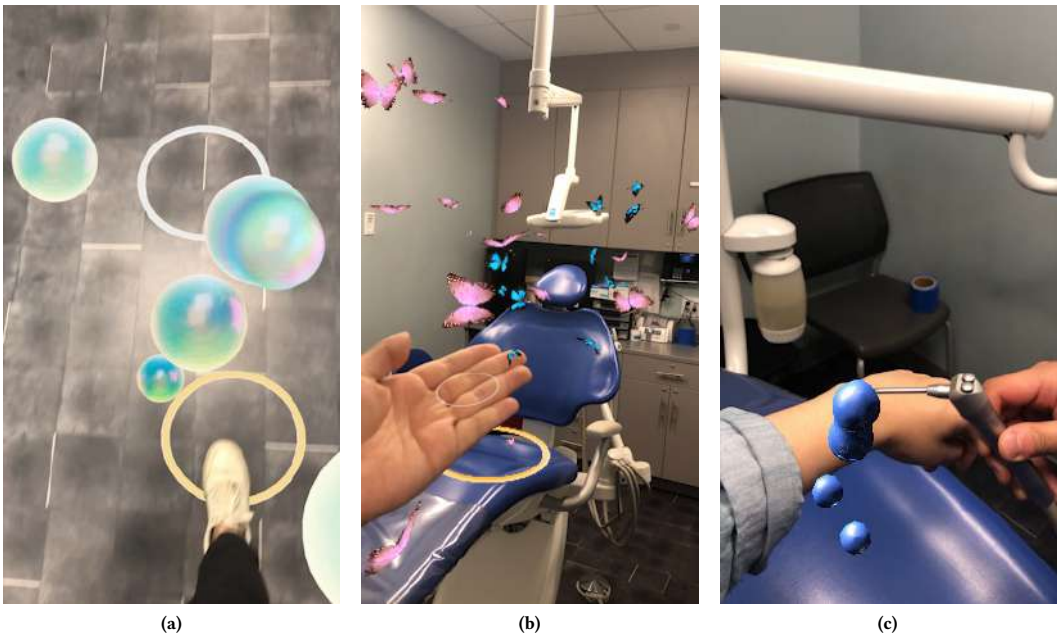


Figure 4: Overview of the AR dental visit system. Image (a) shows the activation of bubble animations upon stepping on a virtual ring placed in the clinic corridor. Image (b) shows the activation of natural elements (e.g., butterflies) animations from a virtual ring positioned on the dental chair, triggered by hand touch. Image (c) shows bubble animations emitting from dental tools upon the detection of a physical marker.

facilitate dynamic updates of these elements by adjusting their rotation and movement to ensure they were correctly positioned in place. Additionally, the system incorporated spatial sounds of slightly higher volume to mask the noises and vibrations from

the dental tools. This multisensory approach created a visually engaging and calming display, and effectively alleviated the stress associated with dental treatment, making users more willing to cooperate with the dentists’ work.

5 EXPERT EVALUATION

During the development of the AR design concept and prototype, we engaged in weekly meetings with OTs, behavioral analysts, dentists, and university healthcare researchers. These interdisciplinary sessions enriched our design process with varied insights and helped mitigate potential risks to patients. Upon completing the initial iteration of our prototype, we conducted the expert evaluation [21] with four OTs, two dentists, and a behavioral analyst, all experienced in working with individuals with severe autism with SPD for over a decade. We opted for expert evaluation over direct user testing for two reasons. First, due to the patients' heightened sensory sensitivities, any inadvertent design shortcomings could potentially distress or harm this vulnerable group. Second, the involvement of patients in the testing phase presents logistical and time-intensive challenges, requiring preparation, clear communication, and meticulous attention to logistical details. Due to the time constraints, we presented the video prototypes to the experts instead of asking them to wear devices on-site.

The consensus among the experts leaned favorably towards the gamification strategy embedded within the AR experience, with an overall agreement on the potential benefits of integrating various sensory elements into the physical environment for the patients. We described the key findings from the expert evaluation and the several critical adjustments in our design below.

- **Bringing nature imagery into the physical environment.**

A behavioral analyst, having over two decades of expertise in assisting individuals with autism, recommended the incorporation of natural elements into the clinical environments to foster calm and relaxation, especially for those with SPD. The analyst envisioned the dental room as a wonderland: *I can imagine the dental chair becoming a hammock between big trees, just like a wonderland.* Motivated by this, we replaced bubbles with butterflies in the dental room, creating an immersive experience just like being in a grassland with natural surroundings.

- **Personalizing the experience based on individual sensory profile.**

Nearly all experts emphasized the importance of the unique sensory sensitivities of each individual. The clinic's practice of collecting sensory profiles from patients' caregivers (Figure 3) was integrated into the prototype. Before the real AR experience, an interface was added for caregivers and occupational therapists to input sensory sensitivity data, enabling customization of the AR experience. This personalization included adjusting the visual intensity of the animations (for example, the number, speed, and colors of bubbles) and the type of soundtracks to suit individual needs.

- **Ensuring empowerment through user control.**

The importance of allowing patients to control their AR experience was highlighted by experts. One dentist remarked: *The most important thing is to not overwhelm them, as they want to have a sense of control over what they see.* Considering this, we incorporated gesture input in the AR experience, giving patients the ability to control the imagery through hand movements.

- **Addressing tactile concerns with AR headset.**

Many experts raised concerns about the comfort of wearing AR glasses, especially for autistic individuals. To address this, we opted for a mobile device-based system over traditional head-mounted headsets like HoloLens. The prototype was developed for mobile devices, using HoloKit [3, 13] as a lightweight holder and adapter to reduce the discomfort that heavier headsets might cause to patients [20].

6 LIMITATIONS AND FUTURE WORK

While this study provided valuable insights into the efficacy of our AR application using a sensory integration strategy in clinical environments for individuals with autism and SPD, it does have several notable limitations. First, given that the investigation into a novel system required initial validation and gathering of expert feedback, our research was limited by the absence of a formal study involving actual users. The extent to which the system can be adapted or customized to meet the wide range of individual sensory profiles and preferences remains unexplored. We plan to conduct comprehensive user studies to examine the effectiveness of the application in future iterations. Second, the expert evaluation was carried out through a video prototype rather than with the actual device. Despite findings from recent research showing a favorable reception towards head-mounted displays among individuals with severe ASD [20], and our study's efforts to address possible discomfort from wearing bulky headsets, the impact of using a lightweight smartphone holder on user comfort and acceptance remains uncertain. Future research is required to identify the most suitable device options for these individuals. Finally, the research was conducted in a specific dental clinic, which may not be generalizable to other clinical environments, as different settings may present unique challenges and opportunities for the implementation of AR-assisted sensory interventions. Further investigation is needed to examine various contextual factors to generalize the findings to meet the diverse needs of such individuals.

ACKNOWLEDGMENTS

We thank the Cornell Pitech Fellowship Program, particularly Deborah Estrin and Malgosia Rejniak, for connecting us with the YAI organization and their generous funding support of our research. We thank the YAI organization, especially Judith Bailey-Hung, Allison Manalastas Valeriano, Abigail Tayamen, and Elizabeth Diviney, and the doctors and occupational therapists at YAI, for their invaluable advice on the research. Lastly, we thank the reviewers for their constructive feedback that improved this paper.

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